

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1757

Roanoke Rapids 138 kV
July 24, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1757. The Interconnection Customer proposes a 68.9 MW battery energy storage facility interconnecting at the Roanoke Rapids 138 kV substation in the North Carolina area. The request was received on May 23, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 580 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,143,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Cedar Falls Wi Substation - North Tower Tap 138kV L	Haggin - North Tower Tap 345kV	134.5	304	6.4
Wing Lane - Seco 138kV	Skaggs Island Jct 2 - Michell Dam 345kV	120.7	413	9.9
Wing Lane - Spring Street 138kV	Mitchell Dam - Skaggs Island Junction 2 Substation 345kV	120.3	309	7.5
Seco - Harvard 138kV Line	Mitchell Dam - Skaggs Island Jct 2 345kV	116.9	524	9.9
Redbed Plains Wind Farm - Fruitridge & S. Land Park 138kV	Mitchell Dam - Skaggs Island Junction 2 345kV Line	113.2	341	6.2
Dp Elroy - N Tower Tap 138kV	North Tower Tap - Mccartney 345kV	112.1	661	6.2
Carlin - Fruitridge & S. Land Park 138kV	Mitchell Dam - Skaggs Island Jct 2 345kV	112.0	385	6.2
Dist 1000 - Birchwood 138kV	Birchwood - Leavitt 345kV	110.8	619	5.1
Haggin 345 kV/138 kV Autotransformer	North Tower Tap - Hagin Substation 345kV	110.8	426	5.3
Igo - Mickens 345kV	Mickens - Newpot News 345kV	110.6	1,089	8.8
Trenton Diesel - Dist 1000 138kV	Leavitt - Birchwood Substation 345kV	110.5	633	5.1
Spring St - Chipola 138kV	Mitchell Dam - Skaggs Island Jct 2 345kV	109.8	300	7.5
Dunn Siding - Birchwood 345kV	Leavitt - Birchwood 345kV	108.5	1,519	7.5
Fredericksburg - Dunn Siding 345kV	Birchwood - Nauvoo 345kV	107.6	1,914	14.9
Redbed Plains Wind Farm - Westcoast Cellufibre 138kV	Mitchell Dam - SkaggsIsland Jct 2 345kV	106.8	528	6.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Kings River - Dg Amp Solar Coldwater 138kV	Skaggs Island Junction 2 - Mitchell Dam 345kV line	106.0	435	5.1
Westcoast Cellufibre - Leavitt 138kV	Mitchell Dam - Skaggs Island Jct 2 Substation 345kV	104.0	469	8.1
Westcoast Cellufibre - Leavitt 138kV #2	Mitchell Dam - Skaggs Island Jct 2 345kV	104.0	469	8.1
Newport News - Mickens 345kV	Mickens - Igo 345kV	102.5	1,268	10.1
Bear Island - Chancellor Row Substation 345kV	Birchwood - Leavitt 345kV	102.2	2,686	24.1
Mccartney Substation - Gaston 345kV	Mccartney - Roanoke Rapids 345kV	102.0	1,659	18.0
North Anna - Dunn Siding 345kV	Dunn Siding - Fredericksburg 345kV	101.9	1,109	13.8
North Tower Tap - Haggin 345kV Line	Mecklenburg Power Sta - Bear 345kV	101.6	1,490	30.1
Fredericksburg - Possum Point 345kV	Birchwood - Nauvoo 345kV	100.9	1,600	7.5
Hull Street Road - Harvard 138kV	Mitchell Dam - Skaggs Island Junction 2 345kV	100.5	661	9.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cedar Falls Wi Substation - North Tower Tap 138kV L	Rebuild line with higher-capacity conductor	\$6,486,000	3% (\$176,000)
Wing Lane - Seco 138kV	Rebuild line with higher-capacity conductor	\$1,766,000	1% (\$16,000)
Wing Lane - Spring St 138kV	Rebuild line with higher-capacity conductor	\$2,776,000	1% (\$32,000)
Seco - Harvard 138kV Line	Rebuild line with higher-capacity conductor	\$3,563,000	1% (\$32,000)
Redbed Plains Wind Farm - Fruitridge & S. Land Park 138kV	Rebuild line with higher-capacity conductor	\$9,229,000	1% (\$64,000)
Dp Elroy - N Tower Tap 138kV	Rebuild line with higher-capacity conductor	\$6,623,000	1% (\$70,000)
Carlin - Fruitridge & S. Land Park 138kV	Rebuild line with higher-capacity conductor	\$1,242,000	1% (\$9,000)
Dist 1000 - Birchwood 138kV	Rebuild line with higher-capacity conductor	\$21,195,000	1% (\$238,000)
Haggin 345 kV/138 kV Autotransformer	Replace transformer with higher-rated unit	\$16,284,000	1% (\$123,000)
Igo - Mickens 345kV	Rebuild line with higher-capacity conductor	\$61,103,000	1% (\$652,000)
Dist 1000 - Trenton Diesel 138kV L	Rebuild line with higher-capacity conductor	\$29,779,000	1% (\$334,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Spring St - Chipola 138kV	Reconductor line	\$3,058,000	1% (\$35,000)
Dunn Siding - Birchwood 345kV	Reconductor line	\$18,793,000	0% (\$102,000)
Fredericksburg - Dunn Siding 345kV	Reconductor line	\$18,792,000	0% (\$101,000)
Redbed Plains Wind Farm - Westcoast Cellufibre 138kV	Reconductor line	\$696,000	1% (\$4,000)
Kings River - Dg Amp Solar Coldwater 138kV	Reconductor line	\$1,154,000	1% (\$7,000)
Westcoast Cellufibre Substation - Leavitt 138kV L	Reconductor line	\$457,000	1% (\$3,000)
Leavitt - Westcoast Cellufibre 138kV Cir. 2	Reconductor line	\$596,000	1% (\$4,000)
Newport News - Mickens 345kV line	Reconductor line	\$22,496,000	1% (\$223,000)
Bear Island - Chancellor Row Substation 345kV	Reconductor line	\$9,873,000	1% (\$55,000)
Mccartney Substation - Gaston 345kV	Reconductor line	\$33,275,000	1% (\$310,000)
Dunn Siding - North Anna 345kV	Reconductor line	\$16,403,000	1% (\$93,000)
N Tower Tap - Haggin Substation 345kV	Reconductor line	\$25,229,000	1% (\$311,000)
Fredericksburg - Possum Point 345kV	Reconductor line	\$27,660,000	0% (\$146,000)
Hull Street Road - Harvard 138kV	Reconductor line	\$375,000	1% (\$3,000)
	Total allocated to project		\$3,143,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.